Representation steering, decodability, and best-of-N generation in a from-scratch protein language model (LIKHA)

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Abstract We study inference-time control and generation quality of **LIKHA**, a 354M-parameter Llama-style protein language model trained from scratch on ~5.5M plant and industrial protein sequences. We reimplement activation steering (Huang et al., *Steering Protein Language Models*, ICML 2025) for LIKHA and run a nine-iteration auto-research loop. Three findings. (1) **Activation steering controls style but not quality.** Steering the residual stream shifts the output *domain* (plant↔industrial) coherently and monotonically, reaching ~35% of the built-in domain tag's effect, but does **not** improve *foldability* (ESMFold confident-fold rate): no direction or magnitude beats the unsteered baseline. (2) **Foldability is decodable but not steerable.** A per-layer linear probe predicts foldability with AUROC 0.98 (vs 0.64 for amino-acid composition, 0.72 for length) — the model clearly *represents* foldability — yet adding that direction does not make it *generate* more foldable sequences. This decode-vs-control gap is the central result. (3) **The practical lever is generation+selection, not steering.** A temperature×top-p grid shows the confident-fold rate rising from 8% (harsh default) to 37.5% at T=0.75, top-p 0.9 — a 4.5× gain from sampling alone — yielding 41 novel, unique, confidently-folding candidate sequences from one cheap run. We ship a canonical generation tool, a reusable foldability scorer (the probe), and a corrected quality figure. Total compute: ~USD 1.3 across nine GPU pods; no weights were modified. Method requirements we discovered: use mean-pool (not last-token) difference vectors, steer a middle band of layers, keep the magnitude small.

Stan Lee V. Natividad · Kooya PH AI Research Technical report, 2026-11-07. Internal. Lane: devops (execution assigned first-hand by the owner). Model and all artifacts are internal; nothing is released and nothing here is experimentally (wet-lab) validated.

1 Motivation

LIKHA generates novel protein sequences. For it to be useful, we want two things: **control** (bias the output toward a desired property) and **quality** (a usable fraction of outputs are real, foldable proteins). Steering / representation engineering promises cheap, inference-time control without retraining. We asked: can we steer LIKHA toward a domain, and — more ambitiously — toward *foldable* proteins, attacking the low confident-fold rate we measured earlier? The honest answer, below, separates what steering can and cannot do, and identifies what actually moves quality.

2 Model and setup

LIKHA stage 1: decoder-only, 28 layers, hidden 1024, 16 heads, SwiGLU MLP (2752), RoPE, RMSNorm, tied embeddings, char-level amino-acid tokenizer + domain tags (`<dom:agri>`, `<dom:industrial>`), context 1024. Trained from scratch, bf16, on `uniref90_plant` (5.08M) + `ph_crop` (0.37M) + `industrial_3a_reviewed_proxy` (0.07M) + `viridiplantae_reviewed` (0.04M); 3 passes; the cosine LR schedule was cut at 87% (an undertrained tail). Export (fp32 sha256 47fc4277…, bf16 82fa0661…) loaded read-only throughout; **no weights were changed in any experiment here**.

Instruments. **ESMFold v1** (audited commit 75a3841e…, weights sha256 2ee07356… pinned, torch ≥ 2.6 `weights_only`) gives a per-sequence mean pLDDT (a computational confidence, not experimental validation). A **domain-NLL gap** metric, `NLL(seq | agri tag) − NLL(seq | industrial tag)`, reads how plant-like vs industrial-like the model considers a sequence (reference: agri tag −0.062, industrial tag +0.052; unconditional baseline −0.051). A fixed **dev/test split** (2,585 / 2,649 of the stage-1 validation sequences) underlies the quantitative claims. All generation runs ran on SECURE RunPod 4090 pods (seconds each) because CPU generation on the dev laptop is impractically slow (the export carries `use_cache=False`).

3 Method

Activation steering, reimplemented from the equations of Huang et al. (their repository is not copied; no clear licence). For an autoregressive PLM, a per-layer steering vector is the difference of means of residual-stream activations between a positive and a negative set. We apply, at each steered decoder layer, $\tilde{h} = h + \alpha \cdot \|h\| \cdot \mathrm{unit}(v_{\mathrm{layer}})$, then rescale $\tilde{h}$ to $\|h\|$, so α is a scale-free fraction of the hidden norm. Three requirements emerged from the loop (Section 4):

1. **Mean-pool, not last-token, difference vectors.** The last-token activation precedes the end token, so its difference vector is contaminated with a "terminate now" direction; adding it collapses generation to length 0–1. Mean-pooling over residue positions removes this.
2. **Steer a middle band of layers** (9–19 of 28), not all layers: steering every layer compounds to collapse even at small magnitudes.
3. **Small magnitude** ($\alpha \approx 0.02$–0.1 of the hidden norm); larger degrades the output away from both domains.

4 Experiments

Nine iterations; each informed the next (an auto-research loop). Raw results in `experiments/bio-behavior-2026-11-05/results/likha_steer_run*.json`, `likha_foldsteer_run2.json`, `likha_bestof_run1.json`, `likha_foldprobe_run1.json`; full log in `docs/LIKHA-SURGERY-LOOP-2026-11-06.md`.

4.1 Domain steering works (iterations 1–5)

Iterations 1–3 (raw last-token vector; all layers; then middle band) all collapsed generation, isolating requirements 1–2 above. With a mean-pool vector on the middle band (iteration 4), coherence is preserved. A sign sweep (iteration 5) gives coherent, monotonic domain control via the domain-NLL gap (all settings 17–22/24 samples terminate properly):

α	domain gap	α	domain gap
+0.05	−0.036	−0.02	−0.027

α	domain gap	α	domain gap
+0.02	−0.050	−0.05	+0.011
0 (baseline)	−0.051	−0.10	+0.019

Negative α moves the output from the plant default (−0.051) toward industrial (+0.019 at −0.10), ~35% of the way to the full-industrial tag reference, coherently. The plant direction is ceiling-limited (the corpus is ~95% plant, so the default is already near-maximally plant). Composition drifts away from both domains as |α| grows (a quality cost). **Conclusion: steering is a real, if modest, inference-time control knob for domain/style**, complementary to (and weaker than) the built-in domain tag.

4.2 Foldability steering fails (iteration 6)

Target: raise the confident-fold rate of unconditional samples. Positive set = foldable (ESMFold pLDDT ≥ 70), negative = unfoldable (< 35); mean-pool vector; middle band; samples folded with ESMFold.

α	frac pLDDT ≥ 70	mean pLDDT
0 (baseline)	0.15	45.3
−0.05	0.05	42.3
−0.10	0.00	33.3
−0.20	0.00	36.3
+0.10	0.05	36.8

No direction or magnitude beats the unsteered baseline; negative α degrades it. **Conclusion: activation steering does not improve protein quality (foldability).**

4.3 Foldability is linearly decodable (iteration 8)

Per-layer L2-logistic-regression probe, 5-fold stratified CV, foldable (n=70) vs unfoldable (n=99). AUROC rises with depth: 0.81 (layer 0) → 0.97 (layer 12) → **0.981 (layer 27)**. Confound baselines: amino-acid composition 0.639, sequence length 0.718 — the deep probe decisively beats both. **The model represents "is this foldable" cleanly and linearly, yet that direction is not a usable generative control (§4.2). This decode-vs-control gap is the report's central finding:** a linear read-out is not a linear write/cause. Practically, the probe is a reusable, ESMFold-free foldability scorer (with a calibration caveat: trained on foldable-vs-garbage, it saturates on in-between generated sequences, so it is a first-pass filter, not a fine ranker).

4.4 Best-of-N sampling is the quality lever (iteration 7)

We had only ever sampled at the harshest setting (T=1.0, top_k=0), which under-sold the model. A temperature×top-p grid, folded with ESMFold:

T	top-p	frac pLDDT ≥ 70	mean pLDDT
0.75	0.9	**0.375**	**63.2**
0.6	0.9	0.333	59.0
0.9	0.9	0.208	53.8

T	top-p	frac pLDDT ≥ 70	mean pLDDT
1.0	0.9	0.125	49.4
1.0	1.0 (old default)	0.083	43.1

The confident-fold rate rises from 8.3% (old default) to 37.5% at T=0.75, top-p 0.9 — a 4.5× gain from **sampling alone**; top-p 0.9 beats 1.0 at every temperature. 41 novel, unique, confidently-folding candidate sequences were saved from this one run. This corrects the earlier "~3% fold" figure as a harsh-sampling artefact.

4.5 Applying the findings (iteration 9)

A canonical tool (`scripts/likha/likha_generate.py`) bakes in the best defaults (T=0.75, top-p 0.9), best-of-N, the foldability probe as a filter, and optional domain steering. The probe and the mean-pool domain vector are saved as reusable artifacts. `docs/LIKHA-GENERATION-CONFIG.md` records the recommended configuration for whenever LIKHA is served.

5 Discussion

The results fall into a clean picture consistent with an external cross-check (a second model independently reached the same diagnosis). **Steering is a style knob, not a quality lever.** Domain is a shallow compositional/stylistic attribute that a residual-stream nudge can shift; foldability is a deep, global, non-linear structural property that is *represented* (decodable at 0.98) but not *controllable* by a single linear direction. The generative bottleneck is the model's distribution — a 354M, undertrained, from-scratch model places limited probability mass on foldable sequences — not a missing steering vector. Steering can redirect a model toward what it already represents in a controllable axis; it cannot manufacture a controllable axis where none exists, nor add missing structural knowledge. The practical wins are therefore (a) better sampling + selection now (best-of-N, +4.5× fold rate, free), and (b) better data / a fold-aware fine-tune later.

6 Limitations

pLDDT is a computational proxy, not experimental validation; every "foldable" here means "computationally predicted foldable." Sample sizes are small (24 per setting; 70/99 for the probe) and baseline fold rates have high variance across seeds (0–15%). Probe labels come from ESMFold, inheriting its biases; the probe saturates (outputs near 0/1) so it is a ranker, not a calibrated probability. **The sharpest challenge to the central claim was tested:** does the probe AUROC hold on ESMFold-labelled *generated* sequences, or is it only an in-distribution (real-vs-garbage) effect? On a fresh batch of 64 best-config samples, the probe reaches **AUROC 0.894** (Spearman 0.63) against ESMFold pLDDT ≥ 70 — it drops from 0.98 but does NOT collapse. So foldability is decodable from generated sequences too, and the decode-vs-control gap is genuine (decodable at 0.89, still not steerable), not merely an evaluation-distribution artefact. A useful corollary: the probe is a usable, ESMFold-free pre-filter on generated candidates (AUROC 0.89), though as a ranker, not a calibrated score. (`experiments/bio-behavior-2026-11-05/results/likha_probe_on_generated.json`.) The best sampling settings produce long sequences that hit the 350-token cap (a low-temperature termination quirk). No wet-lab validation. No cluster-disjoint homology analysis of the candidates yet. One model, one architecture, one corpus.

7 Conclusion and future work

Activation steering on a from-scratch protein LM controls domain/style but not foldability; foldability is linearly decodable but not steerable; and generation+selection (best-of-N with proper sampling) is the real, cheap quality lever (8% → 37.5%). Future work, in priority order (refined against an external review): (1) *done* — the probe was validated on generated sequences (AUROC 0.894), confirming the gap is genuine and the probe is a usable filter; (2) **LIKHA v2** (larger, diverse, deduplicated data, annealed schedule) as the primary lever on quality; (3) a small **LoRA fine-tune with a composite reward** (ESMFold/probe foldability + novelty + sequence plausibility − collapse penalty), reference-model KL-constrained, attention projections first — the recommended way to shift the *generative* distribution (vs a fixed steering direction, which cannot); a rank-1 `down_proj` weight edit is a later interpretability experiment, not an equivalent of activation steering (a rank-1 edit is input-dependent, not an unconditional residual add). Keep domain steering as a reversible inference hook rather than baking it into the immutable base checkpoint. Then a homology-aware novelty analysis (MMseqs2) of the saved candidates, and eventually wet-lab validation of a shortlist with a partner laboratory.

8 Appendix: reproduction

Code: `scripts/fleet/likha_steer_pod/` (pod jobs: `steer_pod.py`, `fold_steer_pod.py`, `bestof_pod.py`, `probe_pod.py`, `apply_pod.py`, with shells and fire scripts). Tool: `scripts/likha/likha_generate.py` + `likha_foldprobe.npz` + `likha_steer_domain_vec_meanpool.pt`. Data: `experiments/bio-behavior-2026-11-05/` (dev/test split, probe labels, all result JSONs, 41 candidates in `likha_bestof_run1.json`). Model export: `D:\PinoAI-data\likha\stage1-step185373\export` (sha-verified). Method source: Huang, Zhu, He, Yao, *Steering Protein Language Models*, ICML 2025 (arXiv 2509.07983), reimplemented from the equations. Nine SECURE 4090 pods, ~USD 1.3 total; no weights modified.